HOMOMORPHISMS OUT OF A FINITELY PRESENTED MODULE COMMUTE WITH FILTERED COLIMITS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of morphisms of modules as the global sections of the sheaf of homomorphisms. Recalled inside the counted proof (source lines 4062-4067). Turns the two sides of the displayed equality into global sections of sheaves of homomorphisms.
- The commutation of the sheaf of homomorphisms out of a finitely presented module with filtered colimits, and the compatibility of global sections with directed colimits under a finiteness hypothesis on the coverings. Both are invoked at the end of the counted proof (source lines 4068-4070) as a pointer to a lemma of this chapter and a cross-chapter pointer. Identify the colimit of the sheaves of homomorphisms with the sheaf of homomorphisms into the colimit and then take global sections.

Source reference. Original lemma lemma-hom-finite-presentation-colimit: sources/stacks-modules/pinned-original-source.tex, source lines 4068–4068. Navigation pointer only: the counted text refers to the Stacks item lemma-hom-finite-presentation-colimit, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference categories-section-posets-limits: sources/stacks-modules/pinned-original-source.tex, source lines 1148–1148. Navigation pointer only: the counted text refers to the Stacks item categories-section-posets-limits, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged

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Source reference. Original reference sheaves-lemma-directed-colimits-sections: sources/stacks-modules/pinned-original-source.tex, source lines 3271–3271. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-directed-colimits-sections, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be a ringed space. Let I be a preordered set and let $(\mathcal{F}_i, \varphi_{ii'})$ be a system over I consisting of sheaves of $\mathcal{O}_X$-modules (see Categories, Section categories-section-posets-limits). Assume*

(1) *I is directed,*
(2) *$\mathcal{G}$ is an $\mathcal{O}_X$-module of finite presentation, and*
(3) *X has a cofinal system of open coverings $\mathcal{U} : X = \bigcup_{j \in J} U_j$ with J finite and $U_j \cap U_{j'}$ quasi-compact for all $j, j' \in J$.*

Then we have

$$\operatorname{colim}_i \operatorname{Hom}_X(\mathcal{G}, \mathcal{F}_i) = \operatorname{Hom}_X(\mathcal{G}, \operatorname{colim}_i \mathcal{F}_i).$$

Proof. Set $\mathcal{H} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{G}, \operatorname{colim} \mathcal{F}_i)$ and $\mathcal{H}_i = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{G}, \mathcal{F}_i)$. Recall that

$$\operatorname{Hom}_X(\mathcal{G}, \mathcal{F}) = \Gamma(X, \mathcal{H}) \quad \text{and} \quad \operatorname{Hom}_X(\mathcal{G}, \mathcal{F}_i) = \Gamma(X, \mathcal{H}_i)$$

by construction. By Lemma lemma-hom-finite-presentation-colimit we have $\mathcal{H} = \operatorname{colim} \mathcal{H}_i$. Thus the lemma follows from Sheaves, Lemma sheaves-lemma-directed-colimits-sections. □